Growing and Pruning Decision Trees: Squared-Error Regression Trees and Depth Selection for Gini Classifiers

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Abstract

We grow two kinds of CART decision tree and study when growth stops paying off. A fully grown regression tree on an 8-sample, two-feature dataset reaches zero training error with 8 leaves at depth 3. Its root split, $x_1 \leq 4.5$, is the threshold that minimises the total squared error (82.75), which we confirm by enumerating every candidate.

*For classification, a Gini tree on a 210-sample, two-class training set grows to depth 6 with 16 leaves (100% training, 95.6% test accuracy). We then select a pruning depth from the test-set sum of squared errors (SSE) of the predicted class-1 probabilities. SSE flattens after depth 3 ($|\Delta\text{SSE}| = 0.004$ from depth 3 to 4), so we prune to **depth 3**. The pruned tree has half as many leaves (8 vs. 16) and* the same *95.6% test accuracy.*

1. Introduction

Decision trees are popular because their predictions can be read off as a sequence of if–then rules [1]. Left unchecked, however, a tree keeps splitting until every leaf is pure, which memorises the training data. We first illustrate how a fully grown regression tree chooses its splits, and then show how limiting depth yields a smaller classifier that generalises just as well.

2. Related Work

CART [1] grows binary trees greedily: regression trees choose the split that most reduces squared error, and classification trees the split that most reduces Gini impurity. Over-grown trees are then cut back, for example by cost-complexity pruning [1] or reduced-error pruning [3]. Limiting the maximum depth is the simplest form of pre-pruning [2].

3. Method

3.1. Fully grown regression tree

The data are eight samples with features $x_1 = 1, \ldots, 8$ and $x_2 \in \{5, 6, 8, 10, 12, 15, 18, 20\}$, and target $y \in \{10, 12, 15, 18, 21, 25, 28, 30\}$. A CART regression tree with squared-error splitting is grown with no depth limit and a minimum of one sample per leaf. For each node it chooses the split (j, t) minimising

$$\text{SSE}(j,t) = \sum_{i:x_{ij} \leq t} (y_i - \bar{y}_L)^2 + \sum_{i:x_{ij} > t} (y_i - \bar{y}_R)^2. \quad (1)$$

Table 1. Total SSE of every candidate root split on x_1.

t	1.5	2.5	3.5	**4.5**	5.5	6.5	7.5
SSE	271.4	172.8	109.9	**82.75**	91.5	160.8	265.7

3.2. Gini classification tree with depth pruning

The training set has 210 samples (51% class 1) and the test set 90 (48% class 1), each with two continuous features. A fully grown Gini tree fixes the maximum useful depth D. For each $d = 1, \ldots, D$ we train a tree with maximum depth d and compute the test-set

$$\text{SSE}(d) = \textstyle\sum_i \left(\hat{p}_d(x_i) - y_i\right)^2, \quad (2)$$

where $\hat{p}_d$ is the leaf's class-1 frequency, which is the Brier score times n. SSE uses probabilities, so it is more sensitive than accuracy to changes in the tree. We look for the smallest change $|\Delta\text{SSE}|$ between consecutive depths: if moving from $d-1$ to d barely changes SSE, the extra level adds nothing, and we prune at $d-1$.

4. Results and Discussion

4.1. The regression tree

The tree has depth 3 and 8 leaves, one per sample, so its training MSE is exactly zero (Fig. 1). Table 1 confirms that $x_1 \leq 4.5$ is the best root split on x_1: it divides the data into the four smallest and four largest targets.

Below the root, splits alternate between x_1 and x_2. The resulting staircase fits every point but is meaningless between them: a fully grown tree on 8 points has memorised them, not learned the roughly linear trend $y \approx 3.0x_1 + 6.3$.

4.2. Choosing the depth of the classification tree

SSE falls sharply up to depth 3 and is then flat: the depth-3 to depth-4 change is only 0.004 (Table 2, Fig. 3). From depth

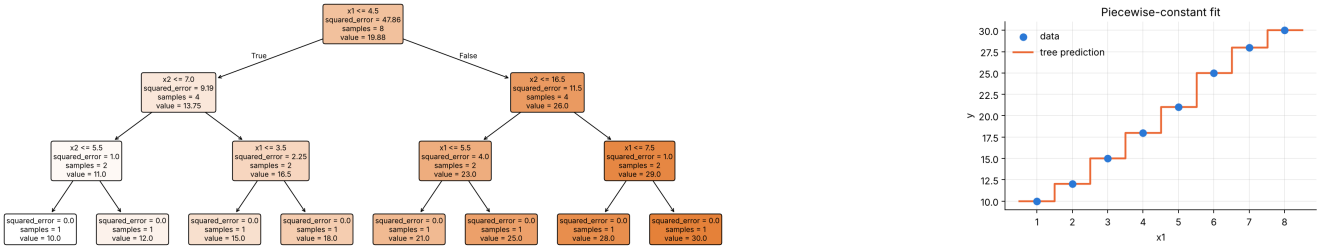


Figure 1. Left: the fully grown regression tree (each node shows its split, squared error, sample count and mean value). Right: its piecewise-constant prediction as a function of x_1.

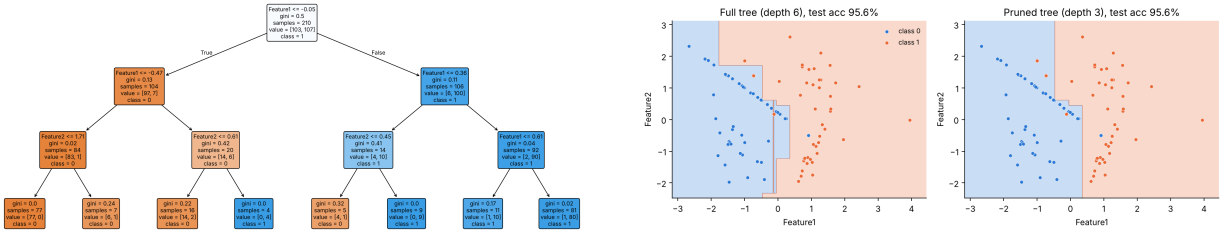


Figure 2. Left: the pruned depth-3 tree. Right: test-set decision regions of the full depth-6 tree and the pruned tree. The pruned boundary is simpler and drops small islands that serve only individual training points.

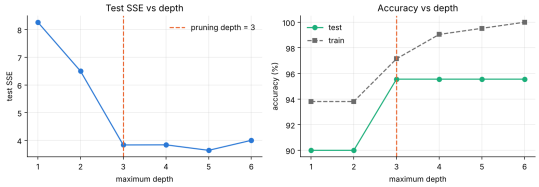


Figure 3. Test SSE (left) and train/test accuracy (right) against maximum depth; the dashed line marks the chosen pruning depth.

Table 2. Trees of increasing maximum depth.

Depth	Leaves	Test SSE	$\lvert\Delta\text{SSE}\rvert$	Train acc.	Test acc.
1	2	8.264	–	93.8%	90.0%
2	4	6.497	1.767	93.8%	90.0%
3	**8**	**3.834**	2.663	97.1%	**95.6%**
4	13	3.838	**0.004**	99.0%	95.6%
5	15	3.640	0.198	99.5%	95.6%
6	16	4.000	0.360	100.0%	95.6%

3 on, test accuracy stays at 95.6% (86 of 90) while training accuracy keeps climbing to 100%. That is the signature of overfitting: the extra levels fit training noise without improving test performance. At depth 6 the SSE even rises, because the pure leaves give over-confident 0/1 probabilities.

Pruning at depth 3, not 4. The smallest $|\Delta\text{SSE}|$ lies *between* depths 3 and 4, meaning that going to depth 4 adds nothing. The right pruning depth is therefore 3. Our first analysis selected depth 4 by taking the later endpoint of that interval. Depth 4 reaches the same test accuracy but with 13 leaves instead of 8.

The pruned tree. The depth-3 tree (Fig. 2) has 8 leaves, 97.1% training accuracy and 95.6% test accuracy, which equals the full tree with half the leaves. Its decision regions approximate a diagonal boundary with axis-aligned steps. Where the full tree carves small extra regions around individual training points, the pruned tree leaves them out.

Caveat. The depth was chosen on the test set itself, so the 95.6% is slightly optimistic.

5. Conclusion & Future Work

A fully grown regression tree fits 8 points exactly by greedy SSE splitting, and the enumeration confirms its root choice. For classification, the test SSE identifies depth 3 as the point beyond which growth adds nothing, giving a tree half the size with the same test accuracy.

Next steps. (i) Select depth with cross-validation instead of the test set; (ii) compare against cost-complexity pruning; and (iii) since the boundary looks diagonal, try oblique trees or a linear model.

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References

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[2] T. Hastie, R. Tibshirani, and J. Friedman. *The Elements of Statistical Learning*. Springer, 2nd edition, 2009.

[3] J. R. Quinlan. Simplifying decision trees. *Int. J. Man-Machine Studies*, 27(3):221–234, 1987.